# M/s HYUNDAI-WIA (INDIA) PVT LTD

# SALES PROPOSAL

# FOR

# TULIP AND OUTER RACE HARDENING AND TEMPERING MACHINE

## QUO-09224-L2D4Y2R2

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# TULIP & OUTER RACE HARDENING AND TEMPERING MACHINE

## APPLICATION:

Induction Hardening & Tempering of bell and stem portions of TULIP and OUTER RACE in a single sequence. The hardness parameter is to be provided by customer. The proposed machine is capable of accommodating Tulips & Outer race only.

## COMPONENTS TO BE ACCOMODATED IN MACHINE:

Machine will be designed to accommodate the range of component mentioned in the RFQ XL document provided “170215_India 4th project”.

## CYCLE TIME:

The cycle time is 25 seconds (FTF).

**Please Note:** The estimated cycle time is calculated theoretically and is subject to trials at our end. Please consider a tolerance of +15% on the above estimated cycle time. Also the cycle time may vary as per the geometry of the component.

**We would like to draw your attention to the following points of our offer:**

1) *“Penalty will be borne by supplier, if supplier delay each steps as approval, design, built machine.”*

   EFD Comment: We accept 0.5% per week delay and to the maximum of 5% of contract value for delivery
   Note:
   Delivery penalty will not be applicable for duration which is attributed to the activity to be carried out by M/s. Hyundai-WIA ., some of the crucial/Major activities are listed below:-
   i) Delay in DAP approval.
   ii) Delay in Advance Payment.
   iii) Delay in providing Samples for trials.
   iv) Delay in Visiting for Inspection and dispatch clearance.

2) *“Work Quality Chart”:*

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| 순 | Control items | Control standard | Remarks |
|---|---|---|---|
| 1 | Surface hardness | - After Queching : 59 ~ 64(HRc)<br>- After Tempering : 58 ~ 64(HRc)<br>(Recommended 58-61 HRC) | Less 2~3 HRC after tempering |
| 2 | O.P.D distortion | - Max : 0.02mm<br>(Serr. Area within 0.01mm top and bottom) | EFD check carefully |
| 3 | Structure | - BELL : ALL Martensite<br>- STEM : Pealite + Ferrite (center Φ3 area) | |
| 4 | Groove distortion | - Top/middle/bottom should be less than 0.03mm | |

EFD Comments: We can reproduce the same results which we have delivered to M/s M/s. Hyundai-WIA (India). We need to discuss regarding the tolerance mentioned for Hardness after Tempering.

3) *"TOOL SIZE and SET UP should be monitoring type".*
EFD Comments: Will provide mandrels to check coils dimensions.

4) *"After tunnel tempering hardness should be HRc 60~61."*
EFD Comments: We cannot accept a 1 HRC tolerance because we might get a measuring error in HV machine close to this.

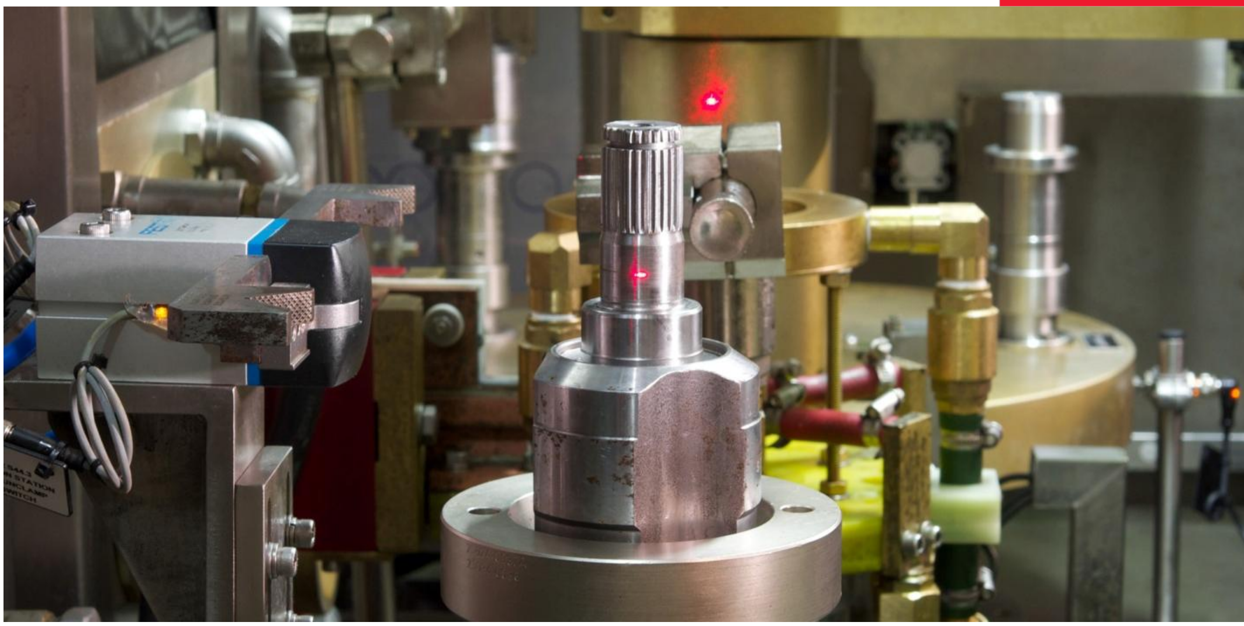
## MACHINE CONFIGURATION:

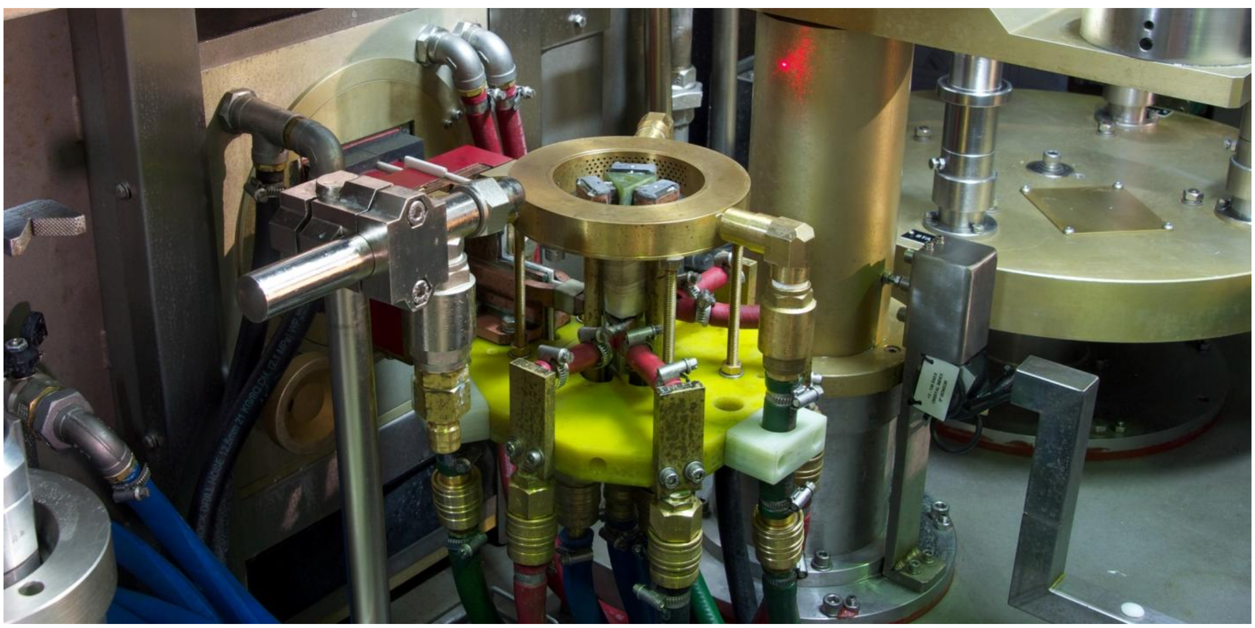
| | | |
|---|---|---|
| Handling Machine | : | LHM With Tunnel Tempering |
| Power source | : | SINAC 300 PM, IGBT Based, Double Output,<br>Maximum Output Power 300 kW,<br>Output Frequency 15-30 kHz for Bell and 5-10 kHz for Stem.<br>25-50 kHz for Female Bore Induction Hardening in Stem area. |
| Quench re-circulating system | : | RQ 350x2-2000-300x2/Suitable7,5 bar or suitable |
| Cooling water re-circulating | : | RC 350-1000-250/Suitable 7,5 bar or suitable system. |
| Process tooling | : | Suitable for 5 Varieties of FOR & 9 Varieties of Tulip. |

## MACHINE DESCRIPTION & PROCESS SEQUENCE:

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inlet Stacker | Washing | Orientation | Bell IH | Stem IH | Tempering | Drying | ECT | Dot matrix Marking | Anti rust application | Out Let Stacker |

Please find herewith some photos of Similar machine for you reference only:-





## SCOPE OF SUPPLY:

| Si.No. | Item / Assembly | Unit | Qty | Technical Specification & make | Price Breakdown |
|---|---|---|---|---|---|
| **1** | **Machine (CNC & Electricals)** | | | | |
| 1.01 | Siemens 840D SL with UPS for CNC | Set | 1 | **SIEMENS 840D SL 4 Axis** | |
| 1.02 | Control & Distribution Panel & Electricals (Rittal Cabinets) | Set | 1 | Rittal Panels | |
| 1.03 | Earth fault detection for Only Hardening Machine | Set | 2 | EFD | |
| 1.04 | Pendants and mounting arm for CNC | Set | 1 | Rittal | |
| 1.05 | Panel air conditioner | Set | 1 | Rittal | |
| 1.06 | Analog module for integration of LVDT | Set | 2 | Suitable | |
| 1.07 | Variable Frequency Drive to vary component rotation speed | No. | 1 | Suitable | |
| 1.08 | Safety light curtain | Set | 1 | Suitable | |
| 1.09 | Color sensors to prevent double hardening in stem station only | Set | 1 | Suitable | |
| 1.1 | Proximity Sensors, Three tier lamp, Poka-yoke, Electrical Controls & Indicators | Set | 1 | IFM / TE | ₹ 1,90,00,000/- |
| | | | | | |
| **2** | **Hardening & Tempering Line (Mechanical, Electrical, Pneumatic & Others) ,** | | | **SPM** | |
| 2.01 | Inlet and Outlet stacker with 200 Nos holding capacity | Set | 2 | EFD | |
| 2.03 | **Component washing unit** with re-circulating water system with filtration, oil skimmer, heater, air blowing and walking beam for job transfer | Set | 1 | EFD | |
| | **Main Hardening machine** | | | | |
| 2.04 | Main Frame (SS Basin) , Machine Housing & Guards (Linear Frame) | Set | 1 | EFD | |
| 2.05 | LVDT for component height checking | Set | 1 | EFD | |
| 2.06 | Component orientation Assembly | Set | 1 | EFD | |
| 2.07 | Z-axis for Gripper (Ball screw / Nut, Steander / L.M Guides, Coupling) | Set | 1 | HIWIN/THK | |
| 2.08 | Y-axis, for Gripper (Ball screw / Nut, Steander / L.M Guides, Coupling) | Set | 1 | HIWIN/THK | |

EFD INDUCTION

| | | | | | |
|---|---|---|---|---|---|
| 2.09 | Z-axis, Suitable for Component up and down movement in stem station (Ball screw / Nut, Steander / L.M Guides, Coupling) | Set | 1 | HIWIN/THK | |
| 2.10 | Y-axis for Feed table for Stem Station | Set | 1 | HIWIN/THK | |
| 2.11 | 3-Jaw Chuck to pick the component from orientation station to heating station | No. | 1 | Jaws/Suitable | |
| 2.12 | Component Intermediate station | No. | | EFD | |
| 2.13 | Additional gripper to transfer the component Intermediate station to Stem Station | No. | 1 | EFD | |
| 2.14 | SEW AC geared motor for component rotation at stem station and rotation assembly | No. | 1 | EFD | |
| 2.15 | 2D table with sealing arrangement for Stem Coil with LVDT for X & Y axis adjustment | No. | 1 | EFD | |
| 2.16 | 2D table with sealing arrangement for Bell Coil with LVTD in X & Y adjustment | No. | 1 | EFD | |
| 2.17 | Feed table for Stem Station with work Piece Lifer | No. | 1 | EFD | |
| 2.18 | Tailstock for stem station | No. | 1 | EFD | |
| 2.19 | Quenching arrangement for stem station | No. | 1 | EFD | |
| 2.20 | Idle station with additional quenching arrangement | Set | 1 | EFD | |
| 2.21 | Cable drag chain for Stem station only | Set | 1 | EFD | |
| 2.23 | Busbar | No. | 2 | EFD | |
| 2.24 | **Automation (Component Handling for Hardening Station)** | | | | |
| 2.25 | Input pick & place arrangement with pneumatic z-axes and x-axes (washing to Orientation station) | No. | 1 | EFD & FESTO | |
| 2.26 | Output pick & place arrangement | No. | 1 | EFD & FESTO | |
| 2.27 | Twin Rejection bin suitable for Tulip with interlock to CNC | Set | 1 | EFD | |
| 2.28 | Manual door, Machine Safety Guarding with safety interlocks and Maintenance Access Door | No. | 1 | EFD | |
| 2.29 | Safety Door Locks | No. | 1 | Suitable | |
| 2.30 | **Tunnel Tempering with Power source SINAC 50 PM 1-3 kHz and Component cooling conveyor** | | | | ₹ 1,00,00,000/- |
| | Controls: SIEMENS PLC | | | | |
| | Pyrometer for temperature measurement | | | | |
| | Handling machine with Heating conveyor and Outlet cooling conveyor for component and Tempering coil. | | | | |
| | **Power Source SINAC 50 PM 1-3 KHz** | | | | |

| | | | | | |
|---|---|---|---|---|---|
| | Cooling re-circulating for Power Source | | | | |
| | Cooling Re-circulating system for Component | | | | |
| 2.31 | **Job transfer to next station from tempering** | | | | |
| 2.32 | **Air blow station before ECT and LASER marking** | No. | 1 | EFD | ₹ 1,02,84,000/- |
| 2.33 | **Eddy current Structural test Station with automation for Bell and stem checking to cover 5 varieties of FOR and 9 varieties of Tulip (Eddy Coils: Stem = 2 Nos., Bell = 20 Nos.)** | No. | 1 | EFD & IBG | |
| 2.34 | **Dot matrix marking Station** | No. | 1 | Marksman | |
| 2.35 | **Pick and place walking beam for transporting the Component for ECT and Dot Matrix marking** | No. | 1 | EFD | |
| 2.36 | **Anti-rust spray unit with, spraying nozzles, chamber and collection tank, Air blowing with Walking beam to transport the component.** | NO. | 1 | EFD | ₹ 10,00,000/- |
| | | | | | |
| **3** | **Machine Accessories** | | | | ₹ 27,00,000/- |
| 3.01 | Pneumatic elements and solenoids | Set | 1 | FESTO | |
| 3.02 | Auto lubrication, circuit and pump, collection trays | No. | 1 | Cenlub | |
| 3.03 | Vented hood to connect fume extractor | No. | 1 | EFD | |
| 3.04 | Mist collector with mounting arrangement one for IH and other for Tempering | No. | 2 | Suitable | |
| 3.05 | Machine painting, labelling | Set | 1 | As per WIA St | |
| 3.06 | SS Compact paper band filter | Set | 1 | SPAN | |
| 3.07 | Tool Rack | Set | 1 | EFD | |
| 3.09 | Air & Quench jet washing guns | Set | 1 | EFD | |
| 3.11 | Rollaway covers and bellow for axis | Set | 3 | EFD | |
| 3.13 | 3-tier lamps, machine indications and controls, machine lamp | Set | 1 | Various | |
| | | | | | |
| **4** | **Power source and accessories** | | | | |
| | Power Source(type, model, frequency, outputs) | | | | |
| **4.01** | **SINAC 300 PM double output, 15-30 kHz for Bell, 5-10 kHz for male stem and for female Bore 25-50 kHz for Female Bore.** | No. | 1 | EFD | ₹ 89,00,000/- |
| 4.02 | Isolation Transformer | No. | 1 | EFD | |
| | | | | | |
| **5** | **Cooling and Quench systems & accessories** | | | | |
| 5.01 | Cooling System | Set | 1 | (RC 350-1000-250)/Suitable | ₹ 22,49,209/- |

| | | | | | |
|---|---|---|---|---|---|
| 5.02 | Quench re-circulating system (Individual pumps & heat exchangers for stem and bell) with oil skimmer | Set | 1 | (RQ 350x2-2000-300x2)/Suitable | |
| 5.04 | Magnetic separator | Set | 1 | Span/Filcon | |
| 5.05 | Booster pump for cooling water circuit (High Pressure) | Set | 1 | Grundfoss | |
| 5.06 | Intermediate tank + Collection tank after filtration with circuit | Set | 1 | EFD | |
| 5.11 | Frame for cooling systems with integral ladder – guarding | Set | 1 | EFD | |
| | | | | | |
| **6** | **Integration & Monitoring** | | | | |
| 6.01 | Integrated piping & instruments | Set | 1 | SS | ₹ 3,00,000/- |
| 6.02 | Electrical Integration & Cabling | Set | 1 | EFD | |
| | | | | | |
| **7** | **Process monitoring system** | | | | |
| 7.01 | Energy Monitoring System | No. | 2 | EFD | ₹ 9,14,400/- |
| 7.02 | Quench Flow Monitoring System | No. | 9 | IFM | |
| 7.03 | Quench Temperature Monitoring System | No. | 1 | EFD | |
| | | | | | |
| **8** | **Process tooling for Tulip (for 5 Varieties FOR and 9 Varieties of Tulip)** | | | | ₹ 30,00,000/- |
| | **Tooling for Five Varieties of FOR** | | | | |
| | Stem Coil | No. | 3 | EFD | |
| | Bell Coil | No. | 5 | EFD | |
| | Fixture on Feed Table | No. | 15 | EFD | |
| | **Tooling for Nine Varieties of Tulip** | | | | |
| | Stem Coil | No. | 6 | EFD | |
| | Bell Coil | No. | 8 | EFD | |
| | Fixture for Orientation | No. | 8 | EFD | |
| | Fixture on Feed Table | No. | 24 | EFD | |
| | Coil Setting Gauges | No. | 22 | EFD | |
| | | | | | |
| **9** | **Process Prove Out and Consistency Test** | | | | |
| 9.01 | Component varieties for tests at EFD | | | 2 variety | |
| 9.02 | Productivity & Consistency tests, number of components | | | 50 Nos.each | |
| 9.03 | Laboratory inspection | | | 10 samples | |
| 9.04 | Customer training | | | 2 days | |
| 9.05 | Machine inspection | | | 2 days | |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| **10** | **Spare Part: Please refer below List** | | | | ₹ 10,52,391 |
| | | | | | |
| **11** | **Installation & Commissioning** | | | | ₹ 3,00,000/- |
| | | | | | |
| **12** | **Packing** | | | | ₹ 3,00,000/- |

**Note:**

1. For cooling re-circulating system, you have to provide cooling water from your cooling tower to the heat exchangers at a temperature range of 28-32°C at a pressure of 4 to 6 bar.
2. Customer to unload the machine and position as per the layout provided by EFD at its site and
3. Provide all the utilities as per the requirements of the machine.

## PRICING:

For items 1.00 through 12.00 listed above **including Packing and loading on to truck at EFD India.**
Fright charges and Transit Insurance to be arranged by customer:-
**Basic Machine price including Packing and Loading on to truck:-**

**₹ 6,00,00,000/-**

**Spare Part as per Sl no:10.00**

| Item No. | Description | Qty | UOM |
|---|---|---|---|
| **CRITICAL SPARE PARTS LIST FOR CAMI 300KW** | | | |
| SICA0008 | CAMI FA PULSMULTI V1 | 1 | NUM |
| SICA0007 | CAMI FA HEXPRO V1 | 1 | NUM |
| SICA0009 | CAMI FA DISTRIB V1 | 1 | NUM |
| PEI00109 | DIODE MODULE | 4 | NUM |
| SIIM0020 | ALCOMIG BOARD | 4 | NUM |
| SIPM0002 | PCB CARD ASSEMBLY-PROTON | 1 | NUM |
| IOT00684 | THERMOSTAT | 1 | NUM |
| IOT00683 | THERMOSTAT | 4 | NUM |
| PEY00025 | TRANSISTOR MODULE | 6 | NUM |
| IOT01384 | PROGRAM ISOLATION AMPLIFIER | 1 | NUM |
| IOT00694 | DISPLAY FOR ISOLATION AMPLIFIER | 1 | NUM |
| SIIM0002 | PCB CARD ASSEMBLY TRANSIG | 1 | NUM |
| PER00066 | SMPS | 1 | NUM |

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| | | | |
|---|---|---|---|
| PER00067 | SMPS | 1 | NUM |
| PER00068 | SMPS | 1 | NUM |
| PER00097 | POWER SUPPLY SMPS FOR CAMI | 1 | NUM |
| PEU00246 | METAL FILM RESISTOR | 4 | NUM |
| PEU00225 | RESISTOR POWER | 4 | NUM |
| IOT00824 | CAPACITOR | 4 | NUM |
| **CRITICAL SPARE PARTS LIST FOR CAMI 50KW** | | | |
| PEI00109 | DIODE MODULE | 1 | NUM |
| SIIM0002 | PCB CARD ASSEMBLY(TRANSI CARD) | 1 | NUM |
| PEI00089 | THYRISTOR | 3 | NUM |
| SICA0008 | CAMI FA PULSMULTI V1 | 1 | NUM |
| SIIM0020 | ALCOMIG BAORD | 1 | NUM |
| PEM00018 | SEMICONDUCTOR FUSE | 5 | NUM |
| IOT00684 | THERMOSTAT (WATER INLET) | 2 | NUM |
| IOT00683 | THERMOSTAT (INVERTOR) | 2 | NUM |
| IOT00683 | THERMOSTAT (PROTON CARD) | 2 | NUM |
| IOT00683 | THERMOSTAT (THYRISTOR) | 2 | NUM |
| PEY00025 | TRANSISTOR MODULE | 2 | NUM |
| PEM00031 | FUSE | 3 | NUM |

**MACHINE FEATURES:**

1. POWER SOURCE MODEL **(SINAC PM)**

The power source offered (SINAC 300 PM) is of transistorized (IGBT) design with peak output power of 300 kW. The power source is a double output version. The output has an operating frequency band is 15-30 kHz for Bell, 5-10 kHz for Stem and 25 -50 KHz for Female tulip bore hardening.

| Item | Qty | Description |
|---|---|---|
| **1** | | **Power Source** |
| 1.1 | 1 | Transistorized Frequency Converter<br>(Parallel compensated)<br>Type Sinac 300 PM, double output<br><br>Max. Output power 300 kW<br>Frequency range for Bell 15-30 kHz<br>Frequency range for Stem 5-10 kHz<br>Frequency Female Tulip Bore Hardening 25-50 KHz<br><br>Features and Modules:<br>• Single output<br>• Nominal output power, 300 kW<br>• Transistorized inverter<br>• Full metering for kW and kHz<br>• Power or voltage control through 4-20 mA signal |
| 1.2 | 2 | Water Cooled Output Tansformer. |
| 1.3 | 2 | Set of inductor connecting brackets, quick clamping type (manual) with groove guide for precise inductor clamping |

## 2. RE-CIRCULATING TYPE COOLING SYSTEMS FOR QUENCH MEDIUM AND COOLING WATER

2.1 1 **Cooling Unit for Power Source, Output Transformer, Busbar and Coils** (heat exchanger principle)
Type: (RC 350-1000-250)/Suitable

Water tank in stainless steel, strainer, circulating pump with motor, plate type heat exchanger, pressure gauges, pressure relief valve, mains water regulation valve, completely pre-assembled on a common frame

Technical Data

| | | |
|---|---|---|
| Cooling capacity | 350 | MJ |
| Tank capacity | 1000 | L |
| Pump flow rate | 250 | L/min. |
| Pump pressure | 7,5 | bar |

2.2 1 **Cooling Unit for Quench Medium**
(Heat exchanger principle)
Type: (RQ 350x2-2000-300x2)/7-5 or suitable

Water tank in stainless steel, Compact paper band filter, with strainers, circulating pump with motor, Pressure gauges, plate type heat exchanger, pressure relief valve, mains water regulation valve, completely pre-assembled on a common frame

Technical Data

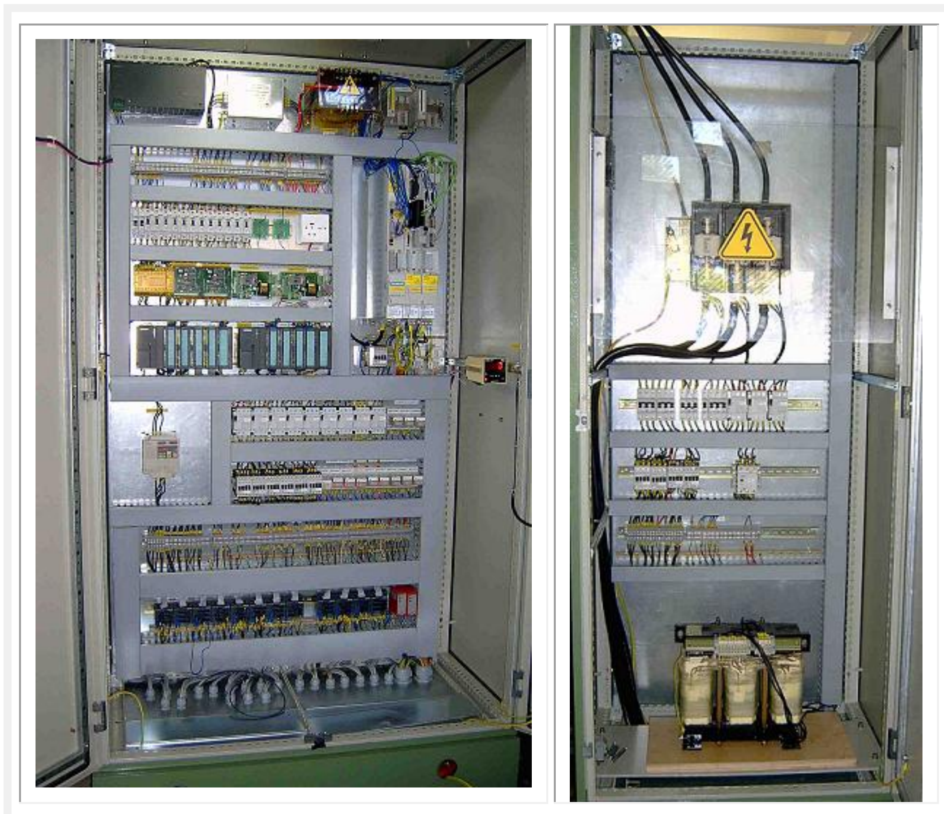
| | | |
|---|---|---|
| Cooling capacity | 350 | MJ X 2 |
| Tank capacity | 2000 | L |
| Pump flow rate for | 300 | L/min X2 |
| Pump pressure | 7,5 | bar |

Please take notice:
The re-circulating systems must be provided with cooling water, max. 32°C, min. 4 to 6 bar pressure, flow rate equivalent to the stated circulating pumps.

### Accessories

2.3 1 Platform to assemble cooling units on top of each other with integral ladder

**3. CONTROL PANEL HOUSING CNC DRIVES, ELECTRICAL SWITCH GEAR AND DISTRIBUTION PANEL HOUSING MAINS AND FUSES**



**4. PNEMATIC CONTROL PANEL**

## ADVANTAGES OF EFD PROPOSAL:

Generator construction in general

EFD-Development centres are constantly working out new, reliable and economic solutions. By exchanging experience with our commercial heat treatment departments as well as commissioning and service staff we are broadening our knowledge in typical production problems. Thousands of generators have been built at EFD. Depending on the application, parallel as well as Parallel compensated converters can be used.

**IGBT parallel compensated transistor generator (SINAC PM)**

The power supply of the series SINAC PM is a parallel compensated frequency converter with diode rectifiers and IGBT transistor inverters. Due to its compact design it requires only minimal space and is still easy to handle. The generator is built from modules that minimise service and maintenance. An auto diagnostic system is integrated into a parameter display simplifying the operation very much. Nominal values (P, U, I and F) are indicated as real values and not in %. The latest parameters are displayed and following that fast and simply controlled. The automatic and electronic matching allows a wide range of applications at nominal power. The output power is continuously adjustable. The working frequency can be determined in a large frequency band according to the application. This gives the possibility of flexible matching ranges to different inductor-designs. The efficiency is high and the creation of harmonic voltages and currents in the network relatively low.

**CNC SYSTEM**

The EFD CNC is an integrated system specially designed to monitor the hardening machine as well as the power source. The advantages can be summarised as follows:

Job Programs: Multiple job programs can be stored which will reduce the human errors while changing the jobs. This also enables quick changeover from one job to another.

Special software: The CNC system incorporates a specially developed software keeping in view the complex process requirements of induction hardening. The CNC monitors all the critical process parameters such as Energy input to the inductor, cooling water to power source/inductor, quench flow and temperature etc. This feature ensures repetition of the same job quality and eliminates the need to repeatedly inspect the processed jobs.

**Error diagnostics :** All the faults that frequently occur in the system like Fuse failure, Emergency stop, Door open, Over temperature, Water failure, Over current etc., are displayed on the monitor. The unique component identification system helps easy and quick access to the failed part.

**User friendly :** Semigraphic alphanumeric backlit LCD display makes programs readable. The programming procedure is extremely user-friendly and is supported by an efficient help menu.

**Flexibility:** Unlimited combinations of Power level, Feed rate and Time delays make the programming very flexible and subsequently the process becomes very easily controllable..

**Documentation :** It is possible to interface with a printer to document the process data.

**Cooling water systems**

EFD cooling water systems have a compact design and are assembled in SS, complete and ready for connection. The stainless plate heat exchangers (counter current principle) have a big cooling surface leading to low water consumption. With the built in supply water regulators the cooling water systems are economical and independent from changes in net water pressure, temperature and quality. Constant pressure, temperature and output conditions guarantee continuous hardening results.

**Construction of inductors and quench assemblies**

EFD has the knowledge and the experience from the delivery of thousands of different types of inductors and quench assemblies collected in a data bank. A group of designers within the construction department is responsible for the process technology and development (including part holding fixture technology). It is company policy that tools are built and tested at EFD according to given construction guidelines, after which we prepare drawings for documentation and reproduction purposes. An express service as well as a repair exchange service guarantee short delivery times for spare parts.

**Test department**

There are several EFD test departments at different sites all over the world. Most of these are equipped with various EFD universal installations and EFD energy sources with different power and frequency. Thus the optimum process technology can be determined even before the placing of an order or during the ordering phase, in close co-operation with the inductor construction department, construction and test lab. Problems can be pointed out in time without causing any delay in delivery time.

**Material technology**

Material labs affiliated with the test departments enable a fast quantification of the test results. For extensive material tests EFD in Freiburg has its own central material lab, equipped with modern preparation and testing technology. In this lab, material and structure can be analysed in quantity and quality. The results of these material tests can be placed at the customer's disposal within the standardised report system.

**Order handling/processing**

Process planning is an integrated part of each negotiation and each order. The product planning computer control ensures a very quick and precise overview from the quotation to the delivery phase. Key dates are controlled at least once a week. The customer's attendance at the internal construction meeting that takes place after the construction phase is highly recommended. In many cases the process is verified with original parts before the pre-acceptance. Each item helps to keep the promised delivery time. We use ERP software MICROSOFT Navision to keep track of material input and resources planning for the project.

**Quality control/quality assurance**

EFD uses basic components from the stock of world-wide leading suppliers. Together with the absolute customer orientation of EFD this guarantees highest operability and endurance of EFD systems for decades.

Defined and controlled processes guarantee a coherent procedure. Responsibilities and contact areas are clearly defined. By means of tests after each operation, beginning with the handling of inquiries down to initiation, errors are minimised. For EFD, quality production means making the customer and his demands on our product the central interest of the company.

We are an ISO 9000 Certified company.

**Pre-acceptance**

The content of the delivery including correspondence with agreed contract is as a rule checked and approved together with the customer during pre acceptance test at our factory. Our project engineer coordinates the pre acceptance test, which guarantees that mechanical and electrical systems as well as process and production conditions corresponds between pre acceptance and the planned location at the customer. Hardening results and structure tests in written form help to carry out the pre-acceptance within the planned period of time.

**Customer Service**

Test and customer service department supervise and co-ordinate the work and operation of initiation and service. A wide range of experienced staff is at your disposal:

- system installation mechanics are experienced mechanical experts for system installation, modifications, overhauling, maintenance and service
- process engineers are trained experts with great general knowledge in the fields of mechanical engineering, electrical engineering, electronics, CNC technology and heat treat technology
- service technicians for power supplies are trained power supply electronics engineers with special knowledge of repair, maintenance, service and calibration of all power supplies sold by EFD as well as power supply adjustment and training

**Training program**

Training is possible at the customer's premises as agreed or at EFD at determined dates. The standard program comprises:

- construction of inductors and quench heads
- basics of induction and power engineering
- preventive care / maintenance
- control technology

**Spare parts**

Spare parts are virtually available via all EFD subsidiaries that have thousands of different standardised commercial articles on stock and are therefore able to deliver within shortest time. Copper material is available from the warehouse supply in all customary sizes and lengths.

**Environmental protection**

The components used are continuously tested for very low pollution effects, if necessary, other solutions are looked for. The implementation of all regulations known to us is documented. Over the medium term, we aim at ecological auditing (DIN 14000).

**Research/Development**

The research of new technologies and process techniques enlarges the knowledge of the EFD personnel. In order to cope with the increasing demands and market developments, EFD has worked out an extensive developmental program for various heat treatment processes for the customer's benefit.

Some examples:

- Pre-heating for laser hardening/melting/welding
- Laser hardening and welding
- Hardening under inert gas atmosphere
- Tempering by induction (instead of conventional)
- Generator Technology

**In addition we profit from the above-average experience with proven solutions and known workpieces and develop them further.**

**The EFD Group**

Take advantage out of an active group. EFD was founded 1996 by the merger of two medium sized traditional companies (FDF and ELVA) and has approximately 1000 employees all over the world today. Manufacturing plants are located in France, Norway, Germany, the USA, the UK, Romania, India and China. Furthermore, there are subsidiaries in Austria, Spain, the UK, Portugal, Sweden, Malaysia, Japan, Thailand and Italy. EFD is the market leader in Europe in terms of delivery of installations for induction surface hardening. The world-wide operating EFD group offers you the co-operation with a competent partner. Active communication and a certified quality system (DIN ISO 9001) guarantee customer-oriented solutions on the highest level. An intensive system of improvement enables continuous technical innovation.

EFD INDUCTION

## COMMERCIAL TERMS:

**1.0 Price Terms:**

- [ ] Ex-works without packing and forwarding
- [x] Ex-works with packing and forwarding
- [ ] Free delivery up to customer's site
- [ ] Free delivery to transport office
- [ ] Free delivery to nearest port

All taxes and duties will be extra at actual and at rates prevailing at the time of despatch.

**2.0 Packing Type:**

- [x] Wooden platform and machine covered with bubble sheet
- [ ] Carton box packing with nylon straps
- [ ] Complete Wooden Box Packing for transport by road / rail
- [ ] Sea-worthy packing

**3.0 Installation & Commissioning Charges:**

- [x] Included
- [ ] Not Included

**4.0 Scope of other charges:**

| | | | |
|---|---|---|---|
| Transportation Charges | [ ] Included | [ ] Not Included | [x] Customer's Scope |
| Transit Insurance | [ ] Included | [ ] Not Included | [x] Customer's Scope |
| Octroi & Others | [ ] Included | [ ] Not Included | [x] Customer's Scope |

**5.0 Delivery Period: Eight Months**

The delivery period is counted from the date of our Order Acceptance. The Order Acceptance will be sent within Five days of EFD receiving the printed copy of the Purchase Order/LOI. The order acceptance is to confirm preliminary acceptance of the order. However, if there are any deviations found in order after detailed contract review it will be addressed in a separate correspondence from EFD. If due to such changes, the delivery date has to be extended it shall be done as per EFD's request.

**6.0 Penalty Clause: -** LD charges 0.5% per week & up to maximum 5% of basic value. Penalty applicable after two weeks of Delivery date.

**Note:**
LD will not be applicable for duration which is attributed to the activity to be carried out by M/s. Hyundai WIA, some of the crucial / Major activities are listed below:-
I) Delay in DAP approval.
ii) Delay in advance Payment.
iii) Delay in providing samples for trials.
iv) Delay in visiting for Inspection and dispatch clearance.

**7.0 Payment Terms:**

30% Advance against corporate Guarantee.
55% Upon Pre-inspection protocol of machine at EFD with Taxes and Duties.
15% Against final acceptance and submission of PBG for 5% of basic Contract value valid for 12 months from the date of acceptance.

**8.0 VALIDITY:** Thirty days from the date of this offer.

**9.0 PRE - ACCEPTANCE TEST:**

Prices include the demonstration of process for maximum of two variety of component. Work pieces must be identical to the parts which will be used for final production at buyer's works. Test pieces to be demonstrated must be available in sufficient quantity at our works at least 12 weeks before the agreed delivery date. If receipt of test pieces is delayed, scheduled delivery may take place without demonstration or written confirmation of the buyer. The same clause regarding delivery will apply if clearance is delayed even after successful process demonstration.

**10.0 FINAL ACCEPTANCE:**

Price includes the demonstration of the process for maximum of 5 variety of FOR and 9 Varieties of Tulip component. Additional prove-outs will be undertaken at extra charges.

**11.0 WARRANTY:**

12 months from the date of Commissioning of 18 Months from the date of Dispatch whichever comes first. Warranty for CNC 18 Months from the date of Commissioning or 24 Months from the date of dispatch whichever comes first.

**12.0 RETENTION OF TITLE:**

Title to all goods will be retained with us till our each and every claim against the buyer originating in the course of the contract has been fully satisfied.

**13.0 COPY RIGHT:**

The specification and technology contained in our proposal is the property of EFD. Passing on this information to third parties as well as unauthorised utilisation by the buyer himself are both not permitted.

**14.0 SUBSEQUENT ALTERATIONS TO ORDER:**

If the order is altered, the price and delivery time may be revised.

**15.0 DISPUTES:**

In accordance with UN 188 A with addendum of 1987.

**16.0 CANCELLATION:**

Actual project costs up to the date of cancellation is to be paid by the customer, plus 18% of total contract value as a compensation for production readjustment costs and loss of project income

***"Any deviations to the above terms are valid only***
***if EFD has accepted in writing prior to order release or during order acceptance."***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